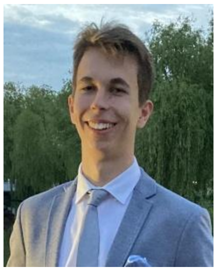
Dániel Sajben



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PERSONAL SUMMARY

Analytical and reliable Computer Science graduate specializing in Artificial Intelligence and Machine Learning. Experienced in building scalable AI and full-stack solutions through academic research and industry internships. Eager to contribute to data-driven innovation and automation projects in a dynamic, international environment.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Java, ML/DL Architectures, JavaScript, TypeScript, HTML/CSS,
- **Frameworks & Tools:** PyTorch, TensorFlow, Keras, Hugging Face, LangChain, LangGraph, Vector Databases, Diffusers, Transformers, AWS, Git, PowerBI, React, Angular
- **Generative AI & LLM Systems**: Retrieval-Augmented Generation (RAG), LLM Fine-tuning (LoRA, QLoRA, instruction tuning), Agentic Systems (OpenAI Agents SDK, Anthropic Agent SDK), Model Context Protocol (MCP), Prompt Engineering & Evaluation, AI Agents, Data Pipelines

EXPERIENCE

Morgan Stanley *Apr. 2025 – Aug. 2025*

- Engineered scalable full-stack solutions using Java Spring Boot and Angular TypeScript.
- Handled large volumes of sensitive financial and user data
- Optimized Sybase SQL queries, reducing average response latency and cutting database load enhancing application responsiveness.

Thesis Research – fMRI Super-Resolution via Generative AI *Jan. 2025 – today*

- Worked alongside a PhD student on a Medical AI research project involving complex fMRI datasets.
- Developed a Latent Diffusion Model (LDM) pipeline to generate synthetic fMRI scans for data augmentation and to super-resolve low-quality images.
- Presented this work at the TDK Student Research Conference in Budapest and won an award.

PROJECTS

Survival Analysis with Tabular Foundation Models *Oct 2025 - Present*

- Developed time-to-event prediction models for medical prognosis utilizing tabular foundation models for feature embedding extraction
- Engineered a unified survival modeling pipeline integrating survival stacking and in-context learning techniques
- Achieved state-of-the-art performance with C-Index of 0.87 on competing risk analysis using SEER dataset (1M+ records) and 0.884 using survival stacking on PBC dataset

Mole Cancer Detection *Sep. 2024 – Jan. 2025*

- Engineered a multi-modal classification system combining CNN transfer learning for dermoscopic image analysis with random forest ensemble for metadata processing across 380,000+ images.
- Achieved 90% accuracy on balanced test dataset through data augmentation strategies and class reweighting techniques and to address severe class imbalance.

Self-driving car *Summer of 2024*

- Built a fully functional self-driving car simulator in JavaScript implementing custom feed-forward neural networks from scratch, achieving 95%+ obstacle detection accuracy without external libraries or frameworks.
- Developed unsupervised learning system that autonomously optimized driving behavior through iterative weight modification and evolutionary selection, utilizing sensor-based linear interpolation for obstacle detection.

Fitness application *Apr. 2024 – Jul. 2024*

- Created a modern fitness app using React + TypeScript and integrated open-source APIs for workout tracking.
- Applied Material UI and UX design principles for a consistent and accessible front-end interface.
- The app is accessible via my personal website under the link provided above in the contact information section.

EDUCATION

University College Dublin **Dublin, Ireland**
Masters in Advanced Artificial Intelligence *Sep. 2026 – Aug. 2027*

Technical University of Munich (QS top 22) **Munich, Germany**
Exchange Semester in Computer Science *Oct. 2025 - March 2026*

- Research in Medical AI
- Research in GNN-LLM Hybrids

Budapest University of Technology and Economics **Budapest, Hungary**
B.S.C. in Computer Science Engineering *Sep. 2022 – Jan. 2026*

- **Concentrations:** Deep Learning and Full-Stack Software Engineering
- **GPA:** 4.61/5.00 (Final grade: Excellent)
- **Related Coursework:** Deep Learning/Machine Learning, Software Architecture, Data Structures & Algorithms, Probability Theory and Statistics

LANGUAGE SKILLS

Spoken Languages

- English C1 (IELTS 8.0/9.0)
- German B1 (Certified)